

Qatar-1b

R-2776

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_250708 · created 2026-09-03T23:36:45+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460864.8076 +/- 0.0032 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.18 +/- 0.054
Transit depth (Rp/Rs)^2	3.25 +/- 1.93 [%]
Orbital Inclination (inc)	81.79 +/- 2.89
Airmass coefficient 1 (a1)	1.23 +/- 0.15
Airmass coefficient 2 (a2)	-0.16 +/- 0.48
Scatter in the residuals of the lightcurve fit is	12.26 %
Best Comparison Star	#1 - [334, 394]
Optimal Aperture	2.01
Optimal Annulus	11.9
Transit Duration (day)	0.049 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.18 ± 0.054 vs 0.1463 (deviation 0.62σ)
Duration fit vs published	0.049 d vs 0.0692 d (deviation 0.75σ)
Mid-time O–C	6.9 min
Depth SNR	3.3
Residual scatter	12.26 %

Light curve

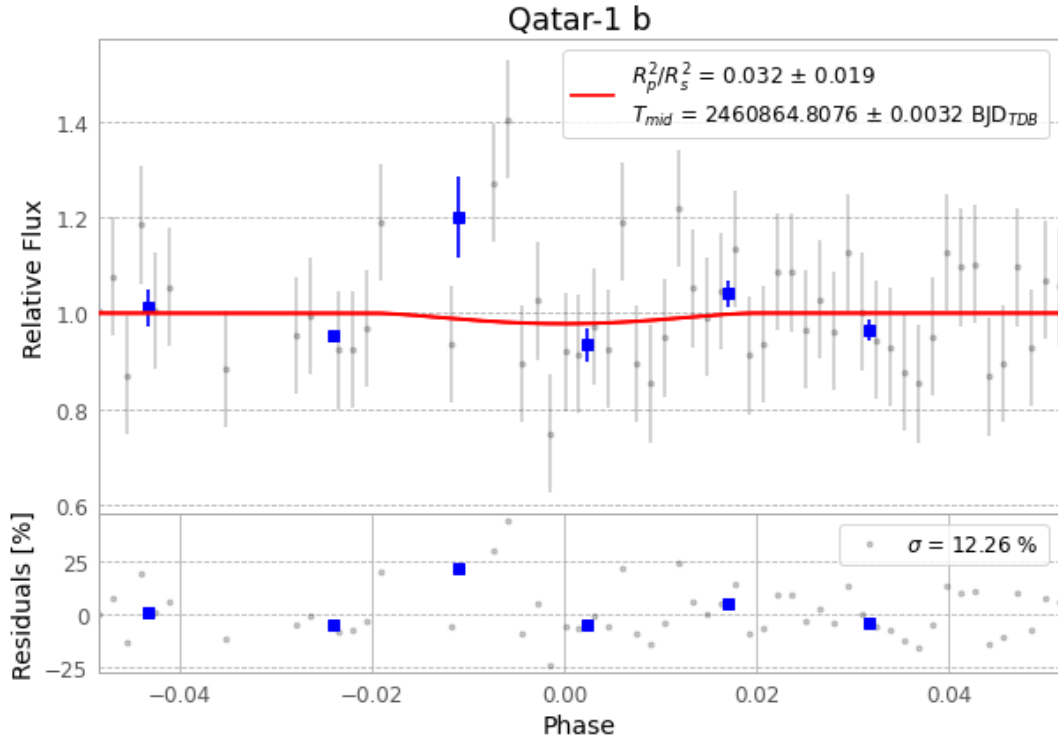


Figure 1 — FinalLightCurve_Qatar-1 b_2025-07-07.png (SHA-256 51344539df81126f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [377, 222], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 c44630e101ba2a5a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

69 FITS frames (45 MB), Qatar-1250708054215.FITS → Qatar-1250708090615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-250708.log (03601d662029...), FinalLightCurve_Qatar-1 b_2025-07-07.png (51344539df81...), FinalLightCurve_Qatar-1 b_2025-07-07.csv (22bceccdc32e...)

Compute resources

Wall time 237.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-03 23:36Z.